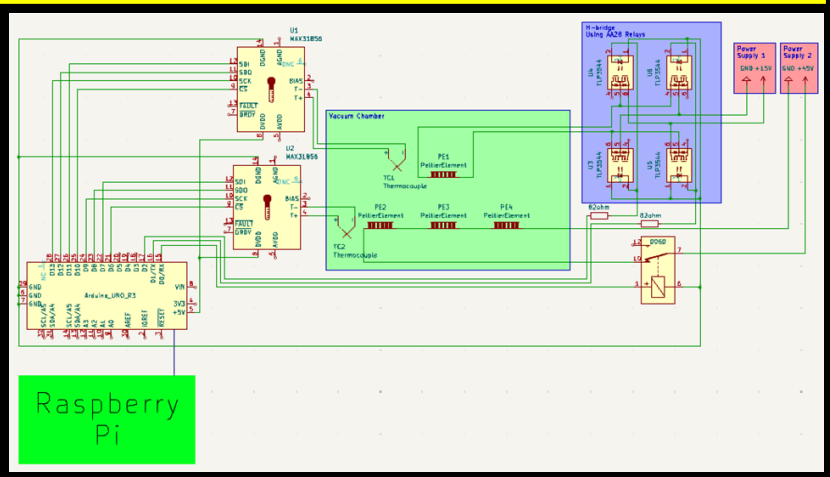
# Design Engineering Portfolio

Oliver Bridge
2026/27

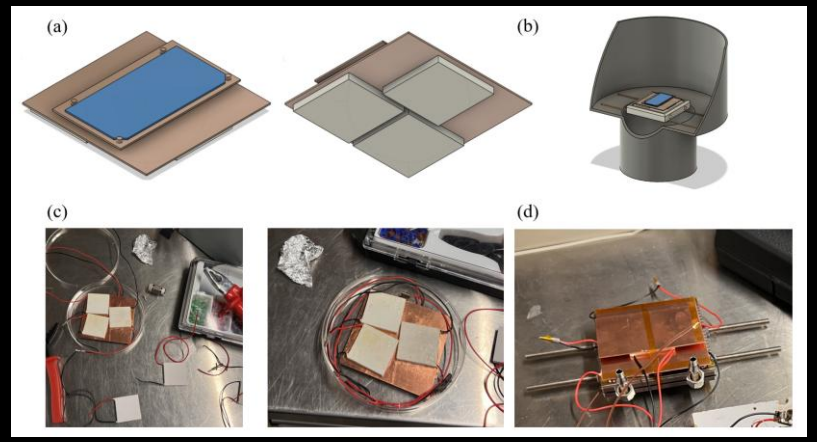
# Thermal Cycling Project

**PROBLEM:** Si solar cells in Low Earth Orbit experience extreme temperature cycles that degrade solder joints over time, but there is limited experimental data on how alternative materials perform under vacuum and thermal stress.

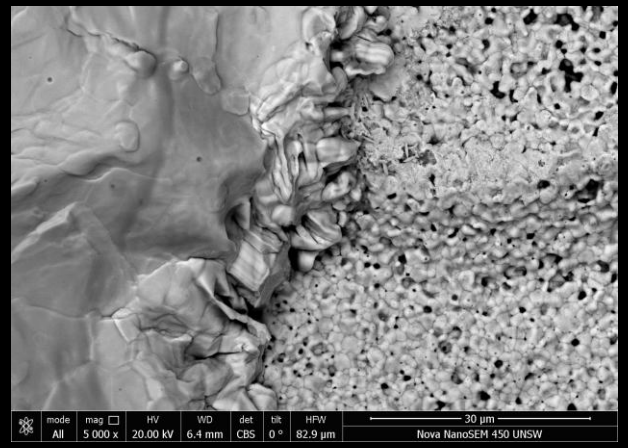
- Converted a disused vacuum deposition chamber into a low-cost thermal vacuum (TVAC) rig simulating Low Earth Orbit
- Designed a water-cooled Peltier stack achieving 1,000 cycles between -50°C and +70°C every 4 minutes (vs ~90 min in orbit)
- Designed a custom control PCB in KiCAD, with Arduino Mega (C++) control and automated Python data logging with Pi
- Validated thermal performance through Ansys and Python simulation
- Tested PERC cells with SnPb and InSn interconnects, analysing degradation with SEM/EDX



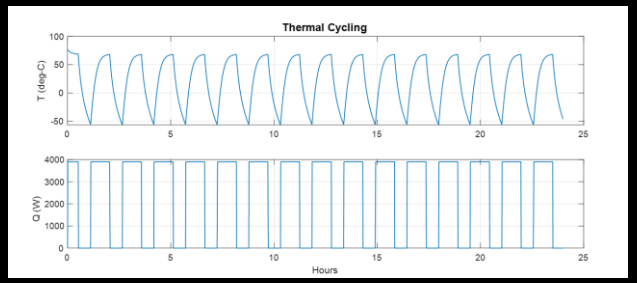

Electrical schematic including thermocouple feedback driving the Peltier stack via H-Bridge




(a) CAD models of thermoelectric stack and, (b) stack inside the chamber (c) Wiring of Peltier devices in series. (d) Full assembly of thermoelectric stack.




SEM showing the boundary between solder and metallisation layer




Thermal cycling data: chamber temperature and Peltier power over time

UNSW, 2025

# Solar Panel metal Recovery project

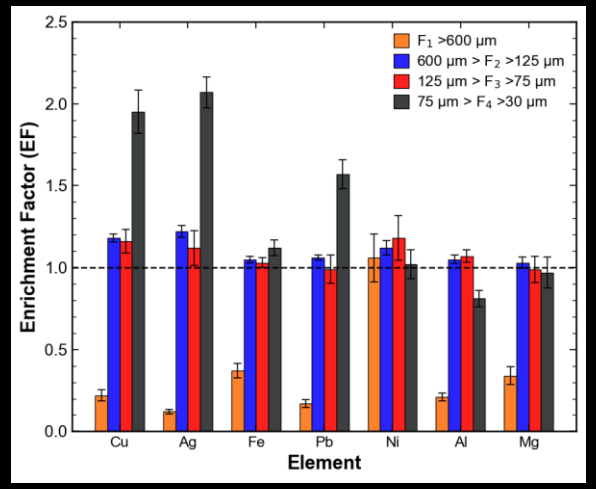
**PROBLEM:** The UK is projected to generate 2 million tonnes of solar panel waste by 2050. Copper, silver and other metals are locked inside the shredded panels, with no sustainable way to recover them, so is currently exported untreated.



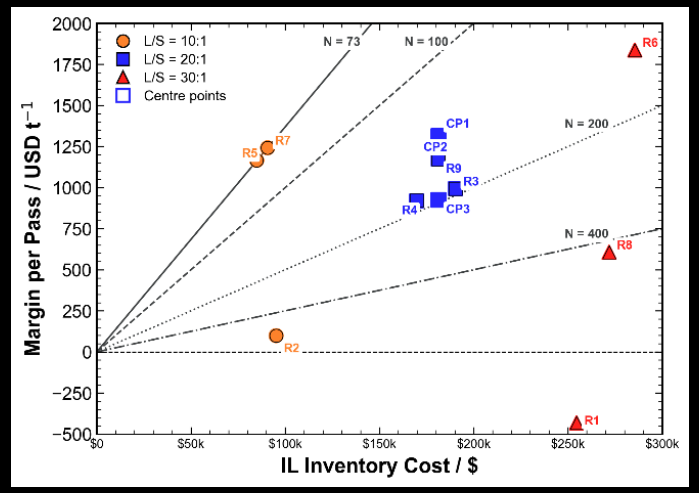
- Worked with the UK's only PV panel recycler characterising real shredded black mass (XRF, ICP-MS, SEM/EDX)
- Screened 5 ionic liquids, then optimised leaching across 4 factors using a Definitive Screening Design ($R^2 = 0.95$)
- Achieved 90.2% copper and up to 69% silver extraction
- Studied electrochemical recovery and identified three failure modes in bulk deposition from real leachate
- Techno-economic assessment showed silver is 98.4% of recoverable value, and the best copper condition was among the worst economically



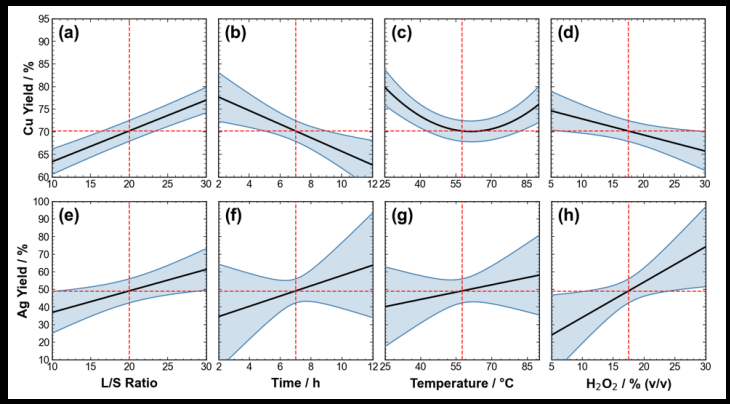
Three electrode cell for copper electrodeposition




Enrichment factor plot by element for post-EL shred sieve fractions




Techno-economic assessment - Ionic liquid recycles needed to break even for each run



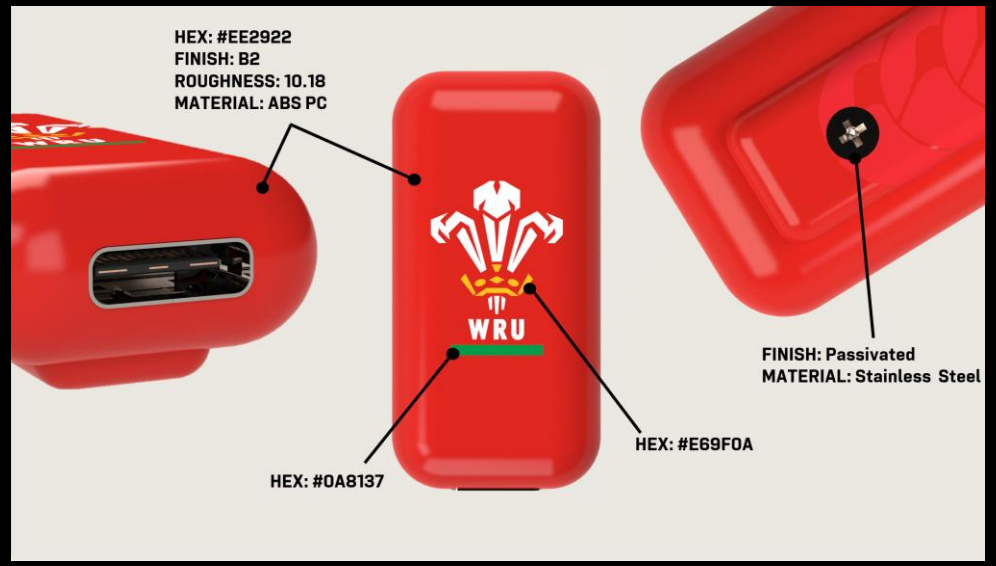

JMP prediction profiler fitted models showing the effect of each leaching factor on Cu and Ag recovery

Imperial, 2026

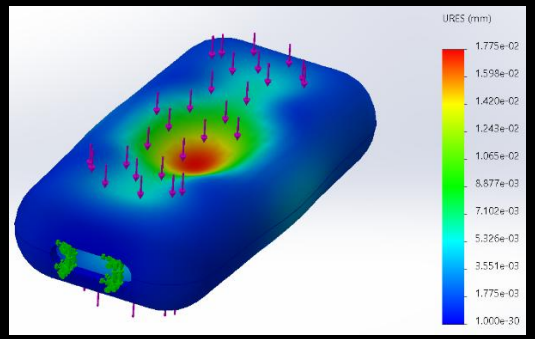
# Head Impact Project

**PROBLEM:** Repetitive, undetected head impacts in contact sports like rugby can lead to long-term brain injury, but existing monitoring tools are expensive, bulky, or inaccessible at amateur level.

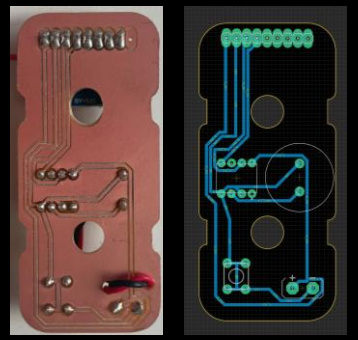
- Designed a low-cost, wearable head impact sensor
- Used FMEA, anthropometric research, real-time motion data
- Developed and designed for rugby with real-world testing
- Optimised design for manufacture of 10,000 units with injection moulds



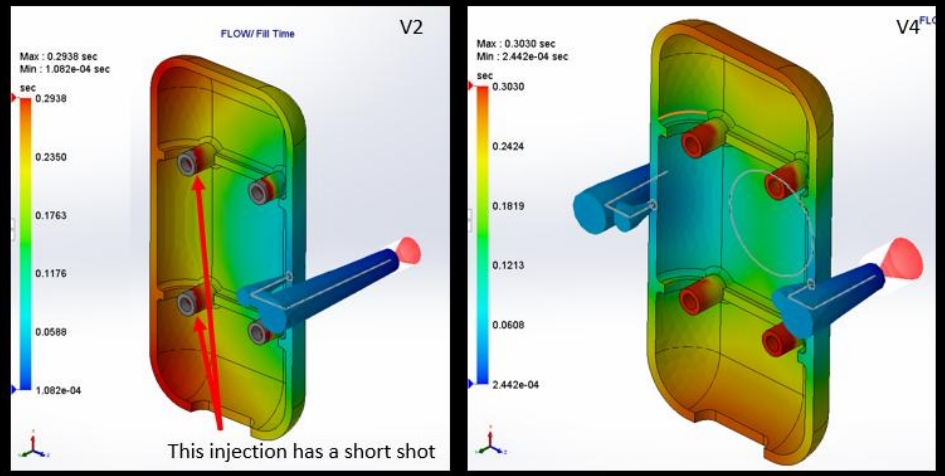

CMF board of the device showing material properties and finishes




FEA simulation showing max displacement on casing with average impact force (1500N)



Prototype PCB with accelerometer, ATmega, actuator and battery

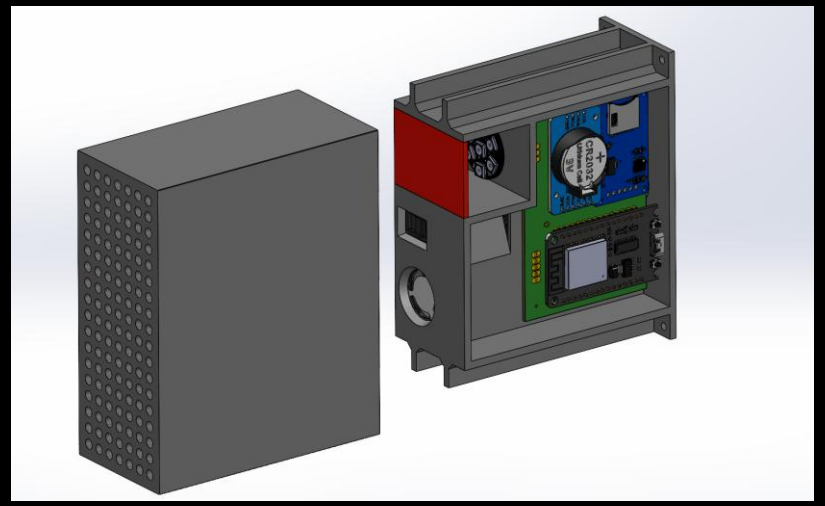



Injection mould simulation showing version iterations

User-Centred Product Design

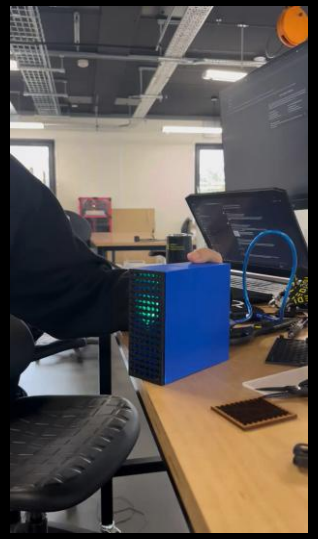
# Air Quality Project

**PROBLEM:** Indoor air pollution is often invisible yet harmful, and current $CO_2$/pollutant monitors do not capture a full picture of pollutant exposure or enable actionable feedback.



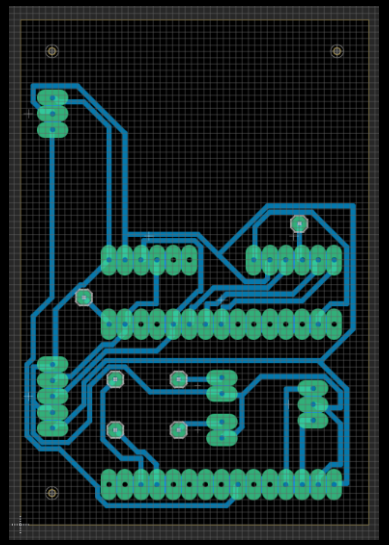
CAD for completed AQ monitor device

- Designed and built an indoor air quality sensor (PM1-10, VOC, NOx)
- Conducted research on detecting pollutants, interface methods and user-feedback
- Created multiple devices and delivered to real communities to guide health decisions






IOS interface



PCB layout design

```
// UK AQI Breakpoints for PM2.5 and PM10
float pm25BreakpointsLow[] = {0.0, 12.0, 24.0, 36.0, 42.0, 48.0, 54.0, 59.0, 65.0, 71.0};
float pm25BreakpointsHigh[] = {11.0, 23.0, 35.0, 41.0, 47.0, 53.0, 58.0, 64.0, 70.0, 500.0};
int pm25AQILow[] = {1, 2, 3, 4, 5, 6, 7, 8, 9};
int pm25AQIHigh[] = {2, 3, 4, 5, 6, 7, 8, 9, 10};
int aqiPm25 = calculateAQI(massConcentrationPm2p5, pm25BreakpointsLow, pm25BreakpointsHigh, pm25AQILow, pm25AQIHigh);

float pm10BreakpointsLow[] = {0, 17, 34, 51, 59, 67, 76, 84, 92, 101};
float pm10BreakpointsHigh[] = {16, 33, 50, 58, 66, 75, 83, 91, 100, 500};
int pm10AQILow[] = {1, 2, 3, 4, 5, 6, 7, 8, 9};
int pm10AQIHigh[] = {2, 3, 4, 5, 6, 7, 8, 9, 10};
int aqiPm10 = calculateAQI(massConcentrationPm10p0, pm10BreakpointsLow, pm10BreakpointsHigh, pm10AQILow, pm10AQIHigh);

// AQI for VOCs and NOx
int aqiVOC = (int)vocIndex * 10;
int aqiNOx = (int)noxIndex * 10;


// Determine the overall AQI
int overallAQI = max(max(aqiPm25, aqiPm10), max(aqiVOC, aqiNOx));

// Determine color
String aqiColor = getAQIColor(overallAQI);

// Set NeoPixel color based on AQI
setNeoPixelColor(overallAQI);
```

AQI function code for pollutant thresholds